Development and Accuracy Analysis of Mobile DGNSS using SatRef Service

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Abstract

Global Navigation Satellite Systems (GNSS) have become crucial to modern positioning and navigation, with smartphones increasingly used as mobile GNSS receivers. However, standalone single-point positioning (SPP) is prone to many error sources, such as ionospheric and tropospheric delays, multipath effect, and noise on the receiver side, which reduce its accuracy and consistency. Differential GNSS (DGNSS) receives pseudorange corrections from reference stations, and applying so can mitigate spatially correlated errors. This leads to better positioning accuracy on mobile devices without the complexities of hardware and processing related to carrier phase.

This project develops and evaluates a mobile DGNSS positioning application for Android, implemented in Kotlin, that receives raw GNSS measurements, decodes GPS navigation messages, and applies real-time pseudorange corrections from the Hong Kong Lands Department Satellite Positioning Reference Station Network (SatRef) via an NTRIP client. The application decodes RTCM version 2 Type 1 correction messages, performs epoch-based weighted least squares position adjustment with 2-sigma outlier rejection, and applies Klobuchar ionospheric model and Saastamoinen tropospheric model with UNBabc mapping function for delay corrections for the SPP reference solution. To evaluate the accuracies of both positioning methods, geodetic benchmarks were surveyed, which serve as the geodetic ground truth. The resolved ECEF coordinates were converted into HK1980 Grid to compare with the control point coordinates. During March and April 2026, a Redmi 14C Android smartphone was used for the field survey, and 22 benchmarks were surveyed across Hong Kong, covering districts such as Sha Tin, Tai Po, Kowloon City, Tsuen Wan, and Sham Shui Po. DGNSS achieved a horizontal mean absolute error (MAE) of 5.26 and a vertical MAE of 4.40m, improving 28.8% and 36.3% respectively, compared to SPP. DGNSS also improved in horizontal precision (decrease in standard deviation of absolute error) by 31.7%.

Overall, DGNSS outperformed SPP in 17 of the 22 surveyed stations (77%). The results confirm the real-time SatRef corrections for smartphone-grade GNSS positioning are practical and effective in Hong Kong, even across urban and suburban environments. In the discussion, this study identifies hardware constraints and multipath could be potential sources of residual errors. Recommendations include multi-constellation support and using dual-frequency receivers for both SPP and DGNSS, and for SPP, using real-time meteorological data for tropospheric modeling.

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Abbreviations

Abbreviation	Full Term
A-GNSS	Assisted Global Navigation Satellite System
BKG	Bundesamt für Kartographie und Geodäsie (Federal Agency for Cartography and Geodesy, Germany)
CS	Control segment
CSV	Comma separated values
DGNSS	Differential Global Navigation Satellite System
DGPS	Differential Global Positioning System
DoD	Department of Defense (United States)
ECEF	Earth-centered, Earth-fixed (coordinate frame)
ENU	East, North, Up (local coordinate frame)
GLONASS	Globalnaya Navigatsionnaya Sputnikovaya Sistema (Russian GNSS)
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
HK1980	Hong Kong 1980 Grid coordinate system
HKSARG	Hong Kong Special Administrative Region Government
ITRF	International Terrestrial Reference Frame
JSON	JavaScript Object Notation
LandsD	Lands Department, HKSARG
MAE	Mean Absolute Error
NLOS	Non-line-of-sight
NTRIP	Networked Transport of RTCM via Internet Protocol
PhiLamH	Latitude, longitude, height (coordinate class in Kotlin code)
PPP	Precise Point Positioning
PRC	Pseudorange correction
PRN	Pseudo-random noise (satellite identifier number)
RMSE	Root mean square error
RRC	Range rate correction
RTCM	Radio Technical Commission for Maritime Services
RTK	Real-time kinematic
SatRef	Satellite Positioning Reference Station Network (Hong Kong)
SDAE	Standard deviation of absolute error
SPP	Single point positioning
SUPL	Secure User Plane Location
TopocentricAED	Azimuth, elevation, distance (coordinate class in Kotlin code)
UDRE	User differential range error
UI	User interface
WGS84	World Geodetic System 1984 (coordinate frame)
Xyz	ECEF x, y, z coordinates (coordinate class in Kotlin code)
ZHD	Zenith hydrostatic delay
ZWD	Zenith wet delay

Chapter 1: Introduction

1.1 GNSS

Global Navigation Satellite Systems (GNSS) provide location and timing information to the users worldwide and have become a crucial part for positioning and navigation in the modern era, especially GNSS-integrated smartphones are on the rise, enabling mobile positioning and location-based services. However, for standard GNSS, there are various error sources such as satellite clock error, ionospheric and tropospheric delays, multipath effects, and receiver noise. To obtain a high accuracy position, these errors must be corrected or mitigated.

1.2 DGNSS

Differential GNSS (DGNSS) is one of the techniques used for improving the positioning accuracy by applying the corrections sent from reference stations with known coordinates, ultimately correcting some errors such as satellite clock errors, and mitigating ionospheric and tropospheric delays. Since some GNSS errors are spatially correlated, it is assumed that close stations have the exact same errors, therefore the errors can be cancelled. Traditionally, DGNSS is often implemented in GNSS receivers to increase their accuracies. With the rise of smartphone usage and the ability to receive raw GNSS messages, it is now feasible and increasingly common to implement DGNSS corrections on mobile devices.

1.3 SatRef

In Hong Kong, the Lands Department (LandsD) provides Satellite Positioning Reference Station Network (SatRef) service, which sends data stream containing real-time correctional data to the users via SatRef Networked Transport of RTCM via Internet Protocol

(NTRIP) server (LandsD, 2025). To integrate DGNSS into mobile devices in Hong Kong, SatRef is often used since the mobile devices are spatially close to the SatRef stations and public access to well-established positioning infrastructure.

This project develops and evaluates a mobile based DGNSS algorithm using raw GNSS measurements and SatRef correction data, analyzes the achieved accuracy, and identifies the key limitations and improvements.

1.4 Project Objectives

Three objectives are defined for this project:

- Develop algorithm

  This project aims to implement the full GNSS positioning workflow into a mobile app using Kotlin. This includes decoding GNSS navigation messages (satellite ephemerides data) and RTCM correction data, handling GNSS measurements and time systems, computing satellite positions from ephemerides, and estimating the receiver position using least squares adjustment. Since Android APIs only cover limited abstractions over navigation, observation data, most of the received binary data needed to be decoded manually.

- Analyze accuracy

  This project also aims to evaluate the positioning accuracy of the resolved location. This involves surveying geodetic control stations with known coordinates, and comparing the positions from standalone GNSS and DGNSS-corrected against the known points, thus the horizontal and vertical accuracy can be obtained for analysis. The analysis can provide insights for identifying the strengths and limitations of DGNSS on mobile device.

- Recommend improvements

With the survey results collected and the limitations identified in the analysis, this project also aims to propose technical and practice improvements. These include enhancements to error correction models, user interface design, change in algorithm and extensibility of the DGNSS positioning software.

Chapter 2: Literature Review

2.1 GNSS

2.1.1 Introduction and principals of GNSS

GNSS serves as space-based tools for finding one's position across the globe. Major satellite system constellation examples include the GPS from the United States, GLONASS from Russia, Galileo from Europe, and BeiDou from China. According to Kaplan and Hegarty (2017), a GNSS system consists of three components, space segment, control segment (CS), and user segment. Space segment refers to the satellite constellation, which is a collection of satellites that orbit the earth and return ranging signals and data messages, for instance satellite ephemeris data back to the users. The control segment maintains the satellites, for instance the satellite health, battery levels, updates the satellite's clock and ephemeris, overall ensuring the satellites can properly function. Lastly the user segment refers to the GPS receivers, which receive the signals transmitted from the satellites and perform the computations, ultimately resolving their own positions.

Positioning relies on measuring the time a signal takes to travel from the satellite to a receiver. Since the signals travel with EM waves, by multiplying the travel time by the speed of light, the receiver can resolve the distance from each satellite. The distance calculated between the satellite and the receiver is the pseudorange, which is an approximate distance

and contain errors that need to be corrected, such as atmospheric delays and clock errors. The ephemeris data can provide the 3-D location relative to the earth's center, and at least four satellites is required to solve for the position of the receiver by solving the least squares solution.

2.1.2 Positioning techniques

The most common and widespread positioning method is single point positioning, or SPP (Rosalie et al., 2017), since post-processing is not required, only uses one receiver and at least 4 satellites, and usually used for resolving one's position instantly. While SPP is operationally simpler, and does not depend on other structures such as a local base station or network connection, SPP usually has more errors and might only be accurate to meter-level. Therefore, it is usually used in coarse positioning where meter level accuracy is acceptable (Santerre et al., 2014).

One of the alternative positioning techniques is precise point positioning (PPP), and can often achieve higher accuracy up to sub-decimeter level. However, according to Afifi and El-Rabbany (2016), one major weakness in PPP is that it takes relatively long time to converge the solution. Due to the accuracy it can provide, many precise satellite equipment has incorporated with PPP positioning technique, however, it is likely infeasible for mobile applications, since the time to resolve ones position is its major drawback.

On the other hand, real time kinematic (RTK) is relatively fast and uses a base station to correct the errors, and can often achieve decimeter level accuracy (Liu et al., 2022). However, the setup is generally more complex and handling carrier phase data. Furthermore, since carrier phase data is precise yet very sensitive, it is generally more susceptible to multipath errors caused by building blockage, and non-line-of-sight (NLOS) can significantly

degrade the signal quality and thus the positional accuracy could be affected (Liu et al., 2022).

Last but not least, differential GNSS (DGNSS) is also an alternative positioning technique that is based on SPP, but also uses the correctional data sent from the base station to the rover station or the receiver, and can often achieve meter-level accuracy (Kaplan and Hegarty, 2017).

2.1.3 Position calculation techniques

In this project, SPP and DGNSS were used. Therefore, we focus on the position calculation techniques for SPP for the basic position before adjustments.

Before resolving the satellite's position, its orbit needs to be known, and the six Keplerian elements define the satellite orbit (Kaplan and Hegarty, 2017). Among the elements, except true anomaly that needs to be calculated, all other values are sent from the satellites in the navigation data.

a	Semi-major axis	define size and shape of orbit
e	Eccentricity	
Ω	Right ascension of ascending node	Orientation of the orbital plane in the inertial system
ω	Argument of perigee	
i	Inclination	
θ	True anomaly	Angle between satellites and the perigee. It denotes the position of the satellite in the orbital plane.

According to Kaplan and Hegarty (2017), after retrieving the ephemeris data containing semi-major axis a and argument of perigee ω, the next steps are to calculate the mean motion (n), mean anomaly (M), eccentric anomaly (E), true anomaly (θ), orbit radius (r), and GPS satellite position (x, y, z) in the orbital plane. They can be calculated by the following equations respectively.

Item	Equation	Equation number
Mean motion (n)	$n = \frac{\sqrt{u}}{(\sqrt{a})^3} + \delta n$	(1)
Mean anomaly (M)	$M_k = M_0 + t_k(n_0 + \delta n)$	(2)
Eccentric anomaly (E)	$E_k = M_k + e \sin(E_k)$	(3)
True anomaly (θ)	$\theta_k = 2 * atan\left(\frac{sqrt(1 + e)}{sqrt(1 - e)} tan\left(\frac{E}{2}\right)\right)$	(4)
Orbit radius of the satellite position (r)	$r = a(1 - e\cos E_k) + \delta r_k$ $\delta r_k = C_{rs}\sin(2\varphi_k) + C_{ic}\cos(2\varphi_k)$ $r_c = r_k + \delta r_k$	(5)
GPS Position in orbital plane	$x_0 = r_c \cos(\varphi_c)$ $y_0 = r_c \sin(\varphi_c)$	(6)
Uncorrected GPS position in ECEF frame	$x = x_0 \cos(\Omega) - y_0 \cos(i) \sin(\Omega)$ $y = x_0 \sin(\Omega) + y_0 \cos(i) \cos(\Omega)$ $z = y_0 \sin(i)$	(7)

At this point, the GPS position in Earth-centered, Earth-fixed (ECEF) coordinate system can be calculated, but the argument of latitude, inclination, orbital radius can be

corrected. The longitude of ascending node can also take into account of the Sagnac effect, which must be corrected before resolving the receiver's location.

Item	Equation	Equation number
Corrected inclination (i)	$i_k = i_0 + \delta i + \delta i_k * t_k$ $\delta i_k = C_{is} \sin(2\varphi_k) + C_{ic} \cos(2\varphi_k)$ $i_c = i_k + \delta i_k$	(8)
Corrected orbital radius (r)	$r = a(1 - e \cos E_k) + \delta r_k$ $\delta r_k = C_{rs} \sin(2\varphi_k) + C_{ic} \cos(2\varphi_k)$ $r_c = r_k + \delta r_k$	(9)
Corrected Argument of latitude	$\delta\varphi_k = C_{us} \sin(2\varphi_k) + C_{uc} \cos(2\varphi_k)$ $\varphi_c = \varphi_k + \delta\varphi_k$	(10)
Longitude of ascending node, corrected for Sagnac effect	$\Omega_k = \Omega_0 + t_k(\delta\Omega - \delta\Omega_e) - \delta\Omega_e * t_0$	(11)
Corrected GPS position in ECEF frame	$x = x_0 \cos(\Omega_k) - y_0 \cos(i_c) \sin(\Omega_k)$ $y = x_0 \sin(\Omega_k) + y_0 \cos(i_c) \cos(\Omega_k)$ $z = y_0 \sin(i_c)$	(12)

The equations for least square adjustments can be described as below:

$$design\ matrix\ B = \begin{bmatrix} \frac{x_0 - x}{\rho} & \frac{y_0 - y}{\rho} & \frac{z_0 - z}{\rho} & 1 \\ \dots & \dots & \dots & \dots \end{bmatrix} \qquad (13)$$

$$error\ function\ f = \begin{bmatrix} pseudorange - \rho_{\text{corrected}} \\ \dots \end{bmatrix} \qquad (14)$$

$$adjustments = (B^T * P * B)^{-1} * B^T * P * f \qquad (15)$$

2.1.4 Position correction techniques

There are many error sources in GNSS positioning, such as satellite clock error, receiver clock error, ionospheric and tropospheric error, multipath error, and noise (Medina et al., 2018). The errors can be described as the following equations:

$$\Phi = \rho + d\rho + c(dt - dT) - I + T + e_{m\varphi} + \lambda N + n_{\varphi} \quad (16)$$

$$P = \rho + d\rho + c(dt - dT) + I + T + e_{mP} + n_P \quad (17)$$

Φ: carrier phase range in m	T: tropospheric error
ρ: pseudorange code range in m	N: ambiguity in carrier phase range (cycles)
$d\rho$: orbit error	$e_{m\varphi}$: multipath error in carrier phase range
dt: receiver clock error	e_{mP}: multipath error in pseudorange
dT: satellite clock error	n_{φ}: noise in carrier phase range
I: ionospheric error	n_P: noise in pseudorange

For single point positioning, satellite clock error is contained within the raw GNSS ephemeris data and can be calculated, while atmospheric errors (ionospheric and tropospheric) can be modeled. This project develops SPP as a reference to DGNSS, and focuses on correcting or mitigating satellite clock error and atmospheric errors, this leaves the receiver clock error being the unknown, which will be solved by using least squares.

2.1.5 Ionospheric delay correction using Klobuchar model

According to Klobuchar (1987), the Earth's ionosphere can greatly affect the velocity of radio waves, slowing down by up to 300ns in extreme cases, this translates to tens of

meters of pseudorange error. The ionospheric delay is dependent on solar activity, time of day, and satellite elevation angle. The Klobuchar model was developed specifically for the GPS L1 frequency (1575 MHz), and has been adopted by GPS to broadcast and correct the ionospheric corrections. According to the GPS SPS Signal Specification by the US Department of Defense (1995), subframe 4 contains the ionospheric parameters. This shows that ionospheric delay correction is an integral part of SPP.

The model first converts the origin latitude and longitude, and elevation into semi circles (Φ_u, λ_u, E), then calculate the Earth's centered angle (ψ), subionospheric latitude and longitude (Φ_I, λ_I), and geomagnetic latitude (Φ_M) in semi circles the equations are given as the following:

Item	Equation	Equation number
Earth's centered angle (ψ)	$\psi = \frac{0.0137}{E + 0.11} - 0.022$	(18)
Subionospheric latitude (Φ_I)	$\Phi_I = \Phi_u + \psi * \cos(\text{azimuth})$ $\Phi_I = \min(\max(\Phi_I, -0.416), 0.416)$	(19)
Subionospheric longitude (λ_I)	$\lambda_I = \lambda_u + \frac{\psi \sin(azimuth)}{\cos(\Phi_I \, in \, radians)}$	(20)
Geomagnetic latitude (Φ_M)	$\Phi_M = \Phi_I + 0.064 * \cos(\lambda_I \, radians - 5.08)$	(21)

After calculating the geomagnetic latitude, we can evaluate the local time (t), amplitude of delay, period of delay, which applies the alpha and beta ionospheric parameters from the ephemeris data, and finally the ionospheric time delay and slant factor.

Item	Equation	Equation number
Local time (t)	$t = \frac{86400}{2\lambda_I} + GPS\ time$ $t = t\ mod\ 86400$	(22)
Amplitude of ionospheric delay	$amplitude = \alpha_0 + a_1\Phi\mathrm{M} + a_2\Phi\mathrm{M}^2 + a_3\Phi\mathrm{M}^3$	(23)
Period of ionospheric delay (x)	$period = \beta_0 + \beta_1\Phi\mathrm{M} + \beta_2\Phi\mathrm{M}^2 + \beta_3\Phi\mathrm{M}^3$	(24)
Phase of delay (x)	$x = \frac{2\pi(t - 50400)}{period}$	(25)
Slant factor (F)	$F = 1 + 16 * (0.53 - E)^3$	(26)
Ionospheric delay (T_{iono})	$T = 5 * 10^{-9} * F, \quad x \geq \frac{\pi}{2}$ $T = (5 * 10^{-9} + \left(1 - \frac{x^2}{2} + \frac{x^4}{24}\right) * amplitude)$ $* F, \quad x < \frac{\pi}{2}$	(27)

To apply the ionospheric delay to the SPP pseudorange, the delay is converted into meters and added to rho (calculated pseudorange).

$$\rho_{iono\ adjusted} = \rho_{\text{iono unadjusted}} + \mathrm{I} \quad (28)$$

2.1.6 Tropospheric delay correction using Saastamoinen model and UNBabc slant delay mapping

According to Saastamoinen (1972) and Davis et al. (1985), as the GNSS signals pass through the atmosphere, the refraction causes the radio waves to slow down, thus the

tropospheric delay. The tropospheric delay model can be separated into hydrostatic (dry) and non-hydrostatic (wet) components. The Saastamoinen model for the zenith hydrostatic delay (ZHD) and zenith wet delay (ZWD) can be expressed with the following equations:

$$ZHD = \frac{0.0022768 * P}{1 - 0.00266 * \cos(2\varphi) - 0.00000028 * \mathrm{H}} \qquad (28)$$

$$ZWD = 0.002277\left(\frac{1255}{T} + 0.05\right)e \qquad (29)$$

where P is the surface pressure (hPa), T is temperature (K), e is water vapor pressure (hPa), H is height in meters, and φ is latitude. For slanted delays, the zenith delays were used together with a mapping function, where ZHD and ZWD each multiplied with a coefficient. According to Guo and Langley (2003), the UNBabc mapping function improved the accuracy for low-elevation satellites down to 2°. It is based on the continued fraction form of the Marini model (Marini, 1972):

$$m(E) = \frac{1 + \dfrac{a}{1 + \dfrac{b}{1 + c}}}{\sin E + \dfrac{a}{\sin E + \dfrac{b}{\sin E + c}}} \qquad (30)$$

In the UNBabc mapping function, value a for hydrostatic and non-hydrostatic required calculation, while b and c values are constants. (Guo and Langley, 2003):

Item	Equation/Expression	Equation number
a hydrostatic (a_h)	$a_h = (1.18972 - 0.026855 * height\ km + 0.10664 \cos \varphi)/1000$	(31)
b hydrostatic (b_h)	0.0035716	
c hydrostatic (c_h)	0.082456	

a non-hydrostatic (a_{nh})	$a_{nh} = (0.61120 - 0.035348 * height\ km - 0.01526 \cos \varphi)/1000$	(32)
b non-hydrostatic (b_{nh})	0.0018576	
c non-hydrostatic (c_{nh})	0.062741	
Slant delay	$slant\ delay = ZHD * m(E)_{hydrostatic} + ZWD * m(E)_{wet}$	(33)

To apply the tropospheric delay to the SPP pseudorange, the delay is converted into meters and added to rho (calculated pseudorange).

$$\rho_{tropo\ adjusted} = \rho_{\text{tropo unadjusted}} + \text{T} \qquad (34)$$

Since the relative humidity data is usually more common to find, the water vapor pressure can be derived from the relative humidity using the Magnus equation, where T is temperature in °C, P is water vapor pressure in kPa, using the saturation vapor pressure over liquid water (Magnus, 1844; Alduchov and Eskridge, 1996).

$$P = 0.61094 * \exp\left(\frac{17.625T}{T + 243.04}\right) \qquad (35)$$

2.2 DGNSS

2.2.1 Introduction about DGNSS

According to Bonnor (2011), DGNSS was one of the first positioning techniques to be widely adopted, commonly used in maritime and aviation navigation. According to Kaplan and Hegarty (2017), differential GPS (DGPS) extends the functionality of a GPS receiver that uses standard positioning service (SPS), by having at least one reference station with known

coordinates. The reference station can compare the measured pseudorange against the theoretical ranges to derive a correction value for each measured satellite, and sent to the end user data that may contain corrections to the receiver's pseudorange measurements, clock, and ephemeris data. Although there are more accurate alternatives such as the RTK, which can provide superior precision, DGNSS remains highly relevant and feasible to implement for mobile applications, due to its implementation being simpler than RTK.

2.2.2 Principals of DGNSS in position correction

The mathematical principals of DGNSS is its nature of cancelling or mitigating errors from two sets of pseudorange equations. According to Liu (2025), after forming the equation for the rover station and receiving the pseudorange correction (PRC) message, the corrected pseudorange can be expressed as follows:

$$\begin{aligned}\mathbf{P^p_{m_corr}} &= \boldsymbol{\rho}^{\mathbf{p}}_{\mathbf{m}} + \cancel{\mathbf{orb^p}} + \mathbf{c}(\mathbf{dt_m} - \cancel{\mathbf{dT^p}}) + \mathbf{I^p_m} + \mathbf{T^p_m} + \mathbf{M^p_m} + \mathbf{n_m} \\ &\quad - \cancel{\mathbf{orb^p}} - \mathbf{c}(\mathbf{dt_b} - \cancel{\mathbf{dT^p}}) - \mathbf{I^p_b} - \mathbf{T^p_b} - \mathbf{M^p_b} - \mathbf{n_b} \\ &= \boldsymbol{\rho}^{\mathbf{p}}_{\mathbf{m}} + \mathbf{c}(\mathbf{dt_m} - \mathbf{dt_b}) + (\mathbf{I^p_m} - \mathbf{I^p_b}) + (\mathbf{T^p_m} - \mathbf{T^p_b}) \\ &\quad + (\mathbf{M^p_m} - \mathbf{M^p_h}) + (\mathbf{n_m} - \mathbf{n_b})\end{aligned}$$

By the first law of Geography (Tobler, 1970), it is assumed that the ionospheric and tropospheric delays are spatially correlated, in other words, the closer the rover and the base station is, the lower the delay differences. To progress forward, we simply assume the errors at the base and mobile stations are the same, thus eliminating the terms in the equation.

$$\begin{aligned}\mathbf{P^p_{m_corr}} &= \boldsymbol{\rho}^{\mathbf{p}}_{\mathbf{m}} + \mathbf{orb^p} + \mathbf{c}(\mathbf{dt_m} - \mathbf{dT^p}) + \mathbf{I^p_m} + \mathbf{T^p_m} + \mathbf{M^p_m} + \mathbf{n_m} \\ &\quad - \mathbf{orb^p} - \mathbf{c}(\mathbf{dt_b} - \mathbf{dT^p}) - \mathbf{I^p_b} - \mathbf{T^p_b} - \mathbf{M^p_b} - \mathbf{n_b} \\ &= \boldsymbol{\rho}^{\mathbf{p}}_{\mathbf{m}} + \mathbf{c}(\mathbf{dt_m} - \mathbf{dt_b}) + \cancel{(\mathbf{I^p_m} - \mathbf{I^p_b})} + \cancel{(\mathbf{T^p_m} - \mathbf{T^p_b})} \\ &\quad + (\mathbf{M^p_m} - \mathbf{M^p_b}) + (\mathbf{n_m} - \mathbf{n_b})\end{aligned}$$

The corrected pseudorange at the mobile station can be expressed as the following equation, with difference in M and n being multipath and noise respectively.

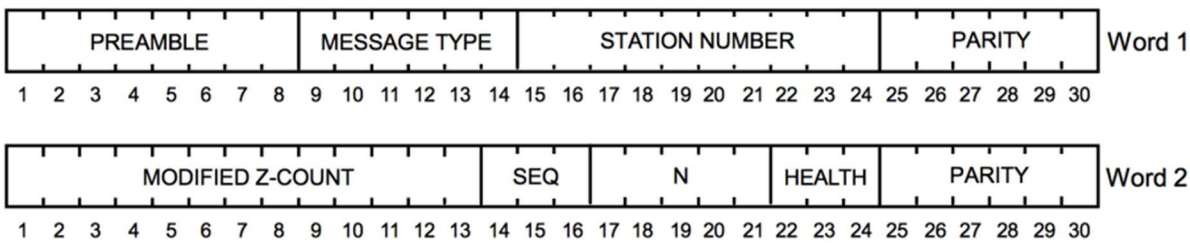
$$\mathbf{P^p_{m_corr} = \rho^p_m + c(dt_m - dt_b) + (M^p_m - M^p_b) + (n_m - n_b)}$$

Since the errors vary across sites and applying error models for general case would be infeasible, they cannot be easily cancelled by making the differential equation. They can be classified as random errors. Other than such errors, the only error left in the equation is the receiver clock error, which can be estimated using least squares adjustments previously mentioned.

2.2.3 Introduction about RTCM data

Radio Technical Commission for Maritime Services (RTCM) is a common data format used for maritime radio communications and navigation. For example, SatRef service sends the correction data to the users using RTCM, with version 3 for RTK and version 2 for DGNSS (LandsD, 2025). According to Lin (2006), RTCM v2 messages consists of 30-bit words. The first two words are the headers while the rest is the body message.

Below shows the two-word header structure (RTCM, 2001).



The preamble marks the start of the header and is always 0x66 (01100110 in binary), the next 6 bits are the message type. The next 10 bits is the station number. In the end of each

30-bit word is the parity bits. The hamming code (32, 26) is applied to the first 24 bits, and the parity bits should match the last 6 bits.

The following table shows the meaning of the RTCM message types (Betke, 2001).

Message type no.	Status according to [1]	Name
1	**Fixed**	**Differential GPS Corrections**
2	Fixed	Delta Differential GPS Corrections
3	**Fixed**	**GPS Reference Station Parameters**
4	Tentative	Reference Station Datum
5	**Fixed**	**GPS Constellation Health**
6	**Fixed**	**GPS Null Frame**
7	**Fixed**	**DGPS Radiobeacon Almanac**
8	Tentative	Pseudolite Almanac
9	**Fixed**	**GPS Partial Correction Set**
10	Reserved	P-Code Differential Corrections
11	Reserved	C/A-Code L1, L2 Delta Corrections
12	Reserved	Pseudolite (Pseudo-Satellite) Station Parameters
13	Tentative	Ground Transmitter Parameters
14	Tentative	GPS Time of Week
15	Tentative	Ionospheric Delay Message
16	**Fixed**	**GPS Special Message**

In the second word, the first 13 bits is the Z count, which is the number of 0.6-seconds in an hour, which gives the reference time information of the base station to the receiver. The value counts from 0 to 6000 and wraps back at zero. The length of frame indicates the number of words of the message, so the receiver can know when to stop when decoding the message. The station health can indicate reliability of the data. For example, 0 indicates reliable, 6 indicates transmission is unmonitored, and 7 indicates the station is not functioning properly.

It should be noted that the sign of the data bits is determined by the parity bits of the previous word (Germany Federal Agency for Cartography and Geodesy, “BKG”, 2008), if the top 2 bits are not 0x00 (00), it indicates the 24 bits of data are bit-inverted. This decoding

process makes the next word reliant on the state of the previous word, which can likely reduce decoding errors.

```
private fun retainOrInvertData(word: Int): Int {
    // The top 2 bits are not 0x00, indicating an inversion
    if ((word and 0x40000000) != 0) {
        // Top 2 bits xor 0, first 24 bits (data) xor 1, last 6 bits (parity) xor 0
        // Xor 1 flips bits, xor 0 retains bits
        return word xor 0x3FFFFFC0
    }
    // Retains all bits
    return word
}
```

Implementation in the Project application, reading subsequent words as 32-bit integer, handling the case where the raw NTRIP data might invert the 24-bit data.

2.2.4 DGNSS correction techniques

DGNSS uses a relatively simple correction technique, by applying the pseudorange correction (PRC) to the uncorrected pseudorange, then perform the least squares adjustment to obtain the adjusted receiver position. The equations can be expressed as the following.

$$P_{dgnss_corrected} = P + PRC + RRC * (t - t0) \qquad (36)$$

Pseudorange correction (PRC) and range rate correction (RRC) are broadcasted from the base station to the receiver, and the magnitude of PRC is much larger than RRC, t is the target time, when the receiver receives the correction data, and t0 is the reference time when the base station receives the satellites, since there is a transmission latency between the base station and the rover, RRC accounts for the latency and applies the correction to the final PRC.

However, sometimes the receiver might receive multiple correction data from different base stations for the same satellite, and we might select the station closest to the

receiver for single base station DGNSS. This would be feasible for long-range DGNSS as the base stations can be far apart, and selecting a single closest one is sufficient. However, in Hong Kong multiple stations would be relatively close to the receiver, in this case we can select the correction data based on the metric. For example, RTKLIB implements a heuristic variance model combining user differential range error (UDRE), station health (k), and latency (Δt) to form a variance of fast correction (Takasu, 2013).

```
*var=varfcorr(p->fcorr.udre)+degfcorr(p->fcorr.ai)*t*t/2.0;
```

$$variance = udre^2 + k * \frac{\Delta t^2}{2} \qquad (37)$$

$$weight = \frac{1}{variance} \qquad (38)$$

Then the correction data with the highest weight can be selected, often indicating high station health, low URDE, and low latency.

2.2.5 Advantages of using DGNSS for mobile devices

For mobile devices, DGNSS offers a balance between accuracy and computational complexity. Since most mobile devices usually use low-cost antennas, and some might not be able to receive the carrier phase data, which is required for RTK. In contrast, DGNSS can operate as long as there is internet to connect to the NTRIP client and receive RTCM data, and the device is equipped with GNSS antenna to receive satellite ephemerides and pseudorange measurements, giving DGNSS a wider smartphone user base. According to Liu (2025), the benefits of using DGPS correction through the Internet is that the range is not limited to line-of-sight distance, not affected by atmospheric attenuation, easy to set up, and low costs.

2.3 SatRef

2.3.1 Introduction about SatRef service

The Satellite Positioning Reference Station Network (SatRef) is a geodetic infrastructure operated and managed by the Lands Department (LandsD) of the Hong Kong SAR Government. This network is equipped with GNSS receivers and monitors the GNSS signals continuously. SatRef is designed to send the correction data in RTCM format to the user through NTRIP. These stations ensures that the general public of Hong Kong can have access to reliable correction data after registration, and its service is free of charge.

2.3.2 Advantages of using SatRef service for mobile devices

Firstly, using SatRef tends to be more accurate than using other DGNSS base stations. By using the SatRef network for mobile DGNSS, the distance between the rover stations and the base stations is reduced and thus the errors can be minimized. In DGNSS positioning, using local base stations is much more preferred than using inter-city ones, due to the ionospheric and tropospheric delays are spatially correlated. By minimizing the delays, the location accuracy can be further improved.

Secondly, since the network is maintained by Hong Kong governmental department, the data quality tends to be high, and the data tends to be more reliable.

Lastly, testing with SatRef is very accessible. Since LandsD provides the data free of charge and only a prompt registration is required, this makes location based app developers, academic researchers, and navigation users easily access the data from SatRef.

Chapter 3: Methodology

This chapter describes the workflow of the mobile DGNSS application development, following the chronological order in which the data is handled and processed within the application. Figure 3A provides an overview of the complete workflow.

PLACEHOLDER: Figure 3A flow diagram

3.1 Application Overview

The application is implemented using Kotlin for Android and features two primary pages: Survey and Survey Store. The Survey page is responsible for all real-time data collection tasks. When a survey is taking place, it shows a real-time count of satellite navigation messages, GNSS observations, and DGNSS corrections received from satellites, including their PRN identifiers and station IDs of SatRef reference stations sending current DGNSS corrections. There is also a button to control the state of the survey - starting and stopping it. After stopping, all collected data are serialized and saved to internal storage.

The Survey Store page is intended for offline data processing and analysis. With the Load button, the user can read the previously saved CSV and JSON files and view a summary of the loaded data: counts of navigation messages, observations, and DGNSS-corrected satellites. After that, pressing the Calculate button performs a full least squares adjustment to find the positions in HK1980 easting, northing, and height coordinates both for SPP and DGNSS algorithms. Additionally, there is a set of benchmark management controls allowing to load a CSV file of Hong Kong geodetic benchmarks. In response, the application finds the closest benchmark and calculates differences in easting, northing, and height coordinates between benchmarks and the results of both SPP and DGNSS algorithms. The

differences of DGNSS algorithm values are marked in color code based on how much better they perform compared to SPP differences - green when DGNSS is better, red when DGNSS is worse and black when both are almost equal.

3.2 Data Collection

3.2.1 GNSS Observations

Raw GNSS pseudorange measurements are gathered using the Android GnssMeasurementsEvent API, which provides the PRN, receiver reception time, and satellite transmission time of each received satellite. Pseudorange is calculated as a difference between the latter two values, multiplied by speed of light. As soon as observations become available, they are appended to an internal list of ObsData objects.

3.2.2 GPS Navigation Message Decoding

Satellite positions are needed to perform the least squares adjustment. The Android GnssNavigationMessage API provides access to raw GPS navigation messages as byte arrays. These byte arrays are parsed manually by extracting Keplerian orbital elements and correction factors, satellite clock parameters and ionospheric correction coefficients from subframe 4. The parsing logic was taken from Google GNSSLogger implementation and re-implemented in Kotlin, with each field's bit position, length and scale factor verified against the IS-GPS-200 specification (US DoD, 1995). Since the set of data for one satellite is spread across several subframes arriving separately via different calls, incoming data are saved first to PartialNavData objects. After all required subframes are received for a certain satellite, they are assembled into a NavData object and added to the list of navigation messages.

3.2.3 A-GNSS Ephemeris via SUPL

In order to make the initial availability of ephemeris data faster, Assisted GNSS (A-GNSS) ephemerides can be obtained from Google's SUPL server (supl.google.com) using Android's location.cts package. Received NavData objects are injected straight into the list of navigation messages to complement or replace entries decoded from subframes.

3.2.4 RTCM v2 DGNSS Corrections

Real-time DGNSS corrections are received using TCP connection established to geodetic.gov.hk, port 2101, mount point DGPS (LandsD, 2026). The connection produces a continuous binary stream compliant with RTCM v2 protocol. RTCM v2 messages consist of 30-bit words, and the last 6 bits of each word carry a Hamming (32, 26) parity bits for error detection. Additionally, it was found that, if D29 and D30 bits of the preceding word are both non-zero, the 24 data bits of the current word must be inverted before usage (RTCM, 1998). Word decoding and parity checking were implemented based on rtcm.c from RTKLIB (Takasu, 2013).

After words have been decoded, and concatenated into a continuous bitstream, Type 1 messages are parsed. Type 1 messages carry pseudorange correction (PRC), range rate correction (RRC), and user differential range error (UDRE) values for each satellite together with the Z-count reference time and station status flags. Type 1 message parsing was implemented based on BKG RTCM2.cpp from RTKLIB (Betke, 2001, and RTCM, 1998). Field bit positions and scale factors were verified accordingly. Decoded corrections are saved into a map with station ID as key, making possible storing corrections received from simultaneously broadcasting multiple SatRef reference stations.

3.3 Data Logging and Reload

At the end of a survey, the application serializes all gathered data into internal storage. Navigation data and GNSS observations are saved as CSV files, while DGNSS corrections are saved as a JSON file. That way, it is possible to reproduce the entire process of positioning and re-run the algorithms indefinitely after saving the data without a need to conduct another field survey. When the Survey Store page is opened, pressing the Load button restores all data back to memory structures.

3.4 Pseudorange Correction

The DGNSS pseudorange correction is applied before the least squares adjustment. For each observation epoch, the application searches the RTCM correction map for a Type 1 record matching the satellite PRN. If there were multiple correction data for the same PRN, then the single optimal RTCM correction will be selected, using a heuristic variance model adapted from RTKLIB (Takasu, 2013), which combines the UDRE, station health indicator k, and transmission latency Δt:

$$variance = udre^2 + k * \frac{\Delta t^2}{2} \qquad (39)$$

$$weight = \frac{1}{variance} \qquad (40)$$

The correction with the highest weight is selected, indicating the correction has low UDRE, good station health, and low latency. The corrected pseudorange is then computed as:

$$P_{dgnss_corrected} = P + PRC + RRC * (t - t0) \qquad (41)$$

where t is the reception time at the rover and t_0 is the Z-count reference time from the RTCM message. The RRC term accounts for the transmission latency between the reference station and the rover, since by the time the receiver gets its own pseudorange, it could be different

from the base station, thus requiring a slightly different adjustment value, and RRC multiplied by the time differences compensates this error. If no correction data are available for a given PRN, that satellite's pseudorange is used uncorrected in the DGNSS solution.

3.5 Least Squares Position Adjustment

3.5.1 Satellite Selection and Filtering

Before the least squares adjustment, two filters are applied to the list of available pseudoranges per epoch. Satellites with an elevation less than 15 degrees are first filtered out since they are particularly sensitive to tropospheric delay, multipath, and line-of-sight obstructions. Additionally, pseudoranges with signal-to-noise ratios below 5 dB-Hz are excluded as they are likely to cause more error than benefit in the solution.

3.5.2 Weight Matrix

The weight matrix is formed as a diagonal matrix for each epoch, with each pseudorange weighted inversely proportional to the squared uncertainty of that pseudorange. The uncertainty is derived based on the pseudorange tracking jitter model developed by Van Dierendonck (1996), where pseudorange noise is directly proportional to carrier-to-noise ratio. Thus, satellites with better signal quality have greater weight in the least squares solution.

3.5.3 Adjustment and Convergence

The least squares solver takes the filtered pseudoranges, navigation data, and approximate receiver location and receiver clock bias from the previous iteration. It calculates the expected satellite positions and satellite clock errors. Using the approximated position from the last iteration, it calculates the expected pseudoranges, then using them to form the design matrix B, which consists of 3 columns of unit vectors for x, y, z, and 1 column of 1s for the receiver clock bias. The error vector f is the difference between measured and computed pseudorange, corrected by atmospheric delays (SPP) and satellite clock error and receiver clock error (both SPP and DGNSS). For SPP, the Klobuchar ionosphere model and the Saastamoinen troposphere model with UNBabc slant delay mapping as discussed in Chapter 2 are used. For DGNSS, pseudoranges are pre-corrected with PRC and RRC as mentioned in Section 3.4, and no receiver-side atmospheric modelling is performed due to spatial correlation cancellation in the differential correction.

The adjustment to the four unknowns is solved iteratively:

$$adjustments = (B^T * P * B)^{-1} * B^T * P * f \qquad (42)$$

The iteration stops when either all four adjustment terms converge to zero, implying that the solution is not converging further, or a maximum number of 100 iterations is exceeded. Once converged, the current position is determined to be the most probable solution at this epoch. The antenna height correction of 1.0 m is then applied to the height coordinate, to account for the phone being held above the benchmark.

3.5.4 Multi-Epoch Averaging and 2-Sigma Rejection

Due to the large receiver clock drift in Android smartphones, and the clock randomly resyncs, each epoch has its own unknown receiver clock bias, and pseudoranges cannot use a single least squares solution. Therefore, position solutions are computed independently for

each epoch and then averaged. However, before averaging, a 2-sigma rejection filter is applied to exclude the positional outliers to ensure the position values agree in general. The remaining positions are then averaged equally to produce the final SPP and DGNSS positions for each survey point.

3.6 Coordinate Transformation to HK1980 Grid

The final averaged position in ECEF is first converted into HK1980 coordinates via seven-parameter Helmert transformation, still in Xyz. Then they are converted into PhiLamH (HK1980 geodetic coordinates) using the Bowring's closed form formula (Bowring, 1976) for faster computation without iterations. Finally, using the parameters to convert from geodetic coordinates to HK1980 Grid provided by the Lands Department (LandsD, 2021; LandsD, 2023). In the app UI, all coordinates are shown in easting, northing, and height using HK1980 Grid system.

3.7 Field Survey

The field surveys were conducted in March and April 2026 using Redmi 14C Android smartphone. The geodetic benchmarks coordinates are from the Lands Department, in the HK1980 grid system. This allows accurate determination of the difference between the survey result and the benchmark coordinates, which were treated as ground truth. Twenty-two benchmarks were surveyed altogether: 11 located in Sha Tin, 3 in Tai Po, 1 in Kowloon City, 3 in Tsuen Wan, and 4 in Sham Shui Po.

During the field survey, the phone was held at approximately chest height in a typical smartphone user posture. After some time spent collecting data, data acquisition was stopped and stored together with the navigation and DGPS correction data. The 1.0 m antenna height

offsets the systematic vertical bias between the phone antenna and the benchmark, to compensate for the surveyor's posture. The in-app calculation of the differences between resolved position and benchmark position (ΔE, ΔN, ΔH) was recorded for further analysis.

3.8 Accuracy Assessment

For all 22 stations, the results in terms of horizontal differences (ΔE, ΔN) and vertical differences (ΔH) were collected in a spreadsheet. Horizontal position error at each station was then computed as:

$$dPos = \sqrt{\Delta E^2 + \Delta N^2} \qquad (43)$$

The following three metrics for horizontal and vertical components, respectively, for both SPP and DGNSS results were calculated: mean absolute error (MAE), root mean square error (RMSE), and standard deviation of absolute error (SADE). MAE is the average absolute value of position errors and serves as the main indicator of positioning accuracy. RMSE puts emphasis on larger error terms and can thus provide additional indication of the performance. SDAE shows the spread of errors between survey sites and acts as the positioning precision indicator.

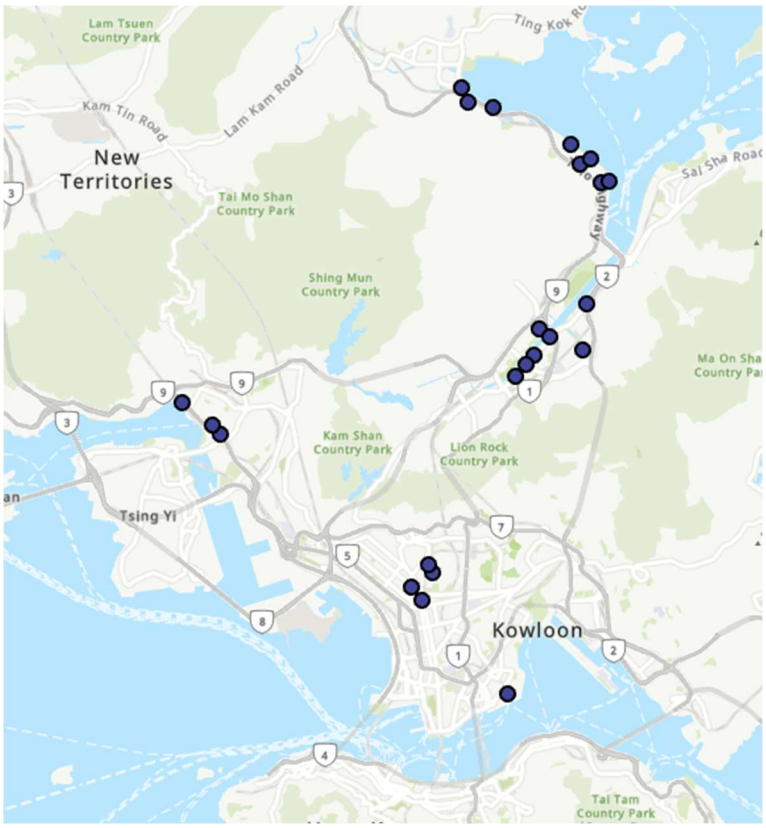
Chapter 4: Results

4.1 Overview of Surveyed Data

A total of 22 Hong Kong geodetic benchmarks were surveyed in March and April 2026. The selected points by district are as follows: 11 stations in Sha Tin, 3 in Tai Po, 1 in Kowloon City, 3 in Tsuen Wan, and 4 in Sham Shui Po (see Table 4B for the full list of

coordinates, station name, and locality details). The field surveys were conducted using Redmi 14C, an android smartphone with raw GNSS receiving capabilities. As mentioned in the methodology, all heights were applied with an antenna height correction of 1.0 meters before averaging with other epochs and outputting the final coordinates, as they were surveyed at chest height like typical smartphone users.

Figure 4A Surveyed point locations



The table below shows the benchmark coordinates from the Lands Department (header in yellow), surveyed SPP coordinates (header in blue), and surveyed DGNSS coordinates (header in green). The recorded survey coordinates were final, averaged coordinates from all epochs, with 2-sigma rejection.

Table 4B Summary of surveyed geodetic control stations

stnName	locality	district	easting	northing	height	easting	northing	height	easting	northing	height
231205	TAI CHUNG KI	SHA TIN	838645	827533	2.9515	838640	827536.6	9.428	838638.1	827533.1	7.79
231206	TAI CHUNG KI	SHA TIN	838238	827083	3.7755	838242.2	827085.6	13.248	838235.4	827084.1	15.781
231207	TAI CHUNG KI	SHA TIN	838017	826841	3.1315	838016.6	826846.6	4.708	838013.6	826844.7	2.544
231208	TAI CHUNG KI	SHA TIN	837740	826529	4.262	837736.9	826536.1	3.749	837737.5	826535.8	7.86
30219	WO CHE ESTA	SHA TIN	838375	827760	5.693	838376.3	827763.2	10.895	838378.1	827759.1	8.872
30003.1	YUEN HONG S	SHA TIN	839488	827217	5.732	839485.7	827218.2	2.383	839489.5	827218.2	4.067
231182	PAK SHEK KO	SHA TIN	840173	831579	6.328	840169.2	831591.8	10.997	840169.6	831588.1	8.481
231183	PAK SHEK KO	SHA TIN	839704	832177	6.639	839702.4	832182.3	0.469	839703.7	832174	4.398
231184	PAK SHEK KO	SHA TIN	839190	832561	6.7735	839189.8	832566	15.08	839192.5	832558.5	5.889
231180	TOLO HIGHW	SHA TIN	838204	833295	6.2605	838205.2	833291.8	17.733	838207.5	833289.8	11.867
231181	TOLO HIGHW	SHA TIN	837490	833326	7.566	837495.1	833336	-4.246	837492.2	833326.7	6.713
200022	TAI PO	TAI PO	837152	833499	5.9235	837147.5	833514.1	17.451	837151.5	833500.8	3.376
200023	TAI PO	TAI PO	836329	834032	5.569	836331.9	834035.3	16.501	836332	834029.7	19.617
231162	NAM WAN RC	TAI PO	835724	834313	4.3605	835724.3	834310.1	-2.362	835722.1	834303.6	2.151
220159.1	SHUNG KING	KOWLOON	837544	818278	4.4225	837548.7	818277.5	3.998	837547.8	818273.6	7.978
221155	TEXACO ROAI	TSUEN WA	830074	825005	4.4955	830077.8	825009.2	-7.356	830078	825004.9	-5.877
220050	MA TAU KOK	TSUEN WA	829885	825253	4.148	829897.4	825254	-4.814	829889.5	825251.2	3.928
220052	HOI KOK STRE	TSUEN WA	829082	825847	4.759	829083.8	825842.1	-6.802	829079.4	825840.7	4.276
220139	PLAYING FIEL	SHAM SHU	835333	820692	5.572	835335.1	820694.6	10.377	835331.2	820693.9	10.617
221207	WONG CHUK	SHAM SHU	835048	821033	5.42	835061.4	821047.3	15.219	835058.4	821043.7	10.063
221187.1	TAI HANG TUN	SHAM SHU	835579	821409	10.062	835582.2	821399	17.529	835582.5	821408.7	5.888
221205	TAI HANG SAI	SHAM SHU	835477	821618	29.365	835468.8	821614	17.552	835473.1	821613.7	27.057

22 coordinates across Hong Kong, using SPP and DGNSS positioning methods.

4.2 Reproducibility and data integrity

The above surveyed coordinates for SPP and DGNSS are fully reproduceable. As for the recorded navigation data, observation data (in CSV format), and DGNSS corrections (in JSON format) are linked in the appendix. The users can reproduce the above coordinates by importing the CSV and JSON files into the `/files/` folder of `data/data/com.example.finalyear`, after installing the app, then import the files after launching the app.

4.3 Single Point Positioning (SPP) results

The standalone SPP solutions were computed using the least-squares adjustment workflow described in the methodology. Horizontal position errors (dPos) ranged from

2.62m to 19.60m, while vertical errors (dH) ranged from -11.85 to +11.53m across the 22 stations.

4.4 DGNSS positioning results

The positioning uses the same base pseudorange, with the addition of SatRef pseudorange corrections, and without modeling the ionosphere and troposphere on the receiver's end, as described in the methodology. dPos ranged from 1.92m to 14.90m, while dH ranged from –10.4m to +14.0m across the 22 stations. Table 4C shows a comparison of horizontal and vertical errors between SPP and DGNSS.

Table 4C Comparison of horizontal and vertical errors between SPP and DGNSS

spp dE	spp dN	spp dP	spp dH	dgnss dE	dgnss dN	dgnss dP	dgnss dH
-5.011	3.592	6.165	6.477	-6.881	0.136	6.882	4.839
4.225	2.626	4.975	9.473	-2.571	1.116	2.803	12.006
-0.441	5.567	5.584	1.577	-3.432	3.697	5.044	-0.588
-3.120	7.087	7.743	-0.513	-2.512	6.764	7.215	3.598
1.336	3.156	3.427	5.202	3.143	-0.897	3.268	3.179
-2.303	1.245	2.618	-3.349	1.487	1.244	1.939	-1.665
-3.841	12.791	13.355	4.669	-3.415	9.105	9.724	2.153
-1.584	5.294	5.526	-6.170	-0.327	-2.992	3.010	-2.241
-0.201	5.045	5.049	8.307	2.515	-2.456	3.515	-0.885
1.216	-3.176	3.401	11.473	3.514	-5.249	6.317	5.607
5.073	9.959	11.177	-11.812	2.214	0.732	2.332	-0.853
-4.470	15.075	15.724	11.528	-0.541	1.840	1.918	-2.548
2.863	3.265	4.342	10.932	3.048	-2.263	3.796	14.048
0.274	-2.921	2.934	-6.723	-1.929	-9.381	9.577	-2.210
4.672	-0.533	4.702	-0.425	3.811	-4.423	5.838	3.556
3.800	4.164	5.637	-11.852	3.967	-0.127	3.969	-10.373
12.354	1.044	12.398	-8.962	4.530	-1.772	4.864	-0.220
1.823	-4.857	5.188	-11.561	-2.592	-6.334	6.844	-0.483
2.064	2.641	3.352	4.805	-1.762	1.902	2.593	5.045
13.409	14.298	19.602	9.799	10.355	10.725	14.908	4.643
3.229	-9.968	10.478	7.467	3.472	-0.289	3.484	-4.174
-8.225	-3.984	9.139	-11.813	-3.896	-4.323	5.820	-2.308

4.5 Accuracy Analysis

4.5.1 Overall Accuracy Summary

Table 4D shows the statistical metrics for the entire dataset, including accuracy metrics mean absolute error (MAE) and root mean square error (RMSE), and precision metric standard deviation of absolute error (SDAE).

Table 4D Overall accuracy summary table (SPP vs DGNSS)

Statistic	SPP (m)	DGNSS (m)	Improvement (%)
Horizontal			
MAE	7.39	5.26	**28.8**
RMSE	8.64	6.08	29.6
SDAE (precision)	4.58	3.13	31.7
DGNSS lower |dPos|	-	17/22	77% of points
Vertical			
MAE	6.90	4.40	**36.3**
RMSE	8.36	5.37	35.8
SDAE (precision)	3.80	3.71	2.3
No. of points	22	22	-
DGNSS lower |dH|	-	17/22	77% of points

The mean absolute error (MAE) in horizontal position was 7.39m and the root mean square error (RMSE) was 8.64m. The vertical accuracy was similar, with MAE of 6.90m and RMSE of 8.36m. DGNSS has achieved a horizontal MAE of 5.26m, (28.8% improvement over SPP), and a vertical MAE of 4.40m (36.3% improvement). The RMSE values show similar results, showing DGNSS has improved 29.6% horizontally, and 35.8% vertically. Among the 22 surveyed stations, DGNSS has outperformed SPP in 17 stations (77%). This confirms the effectiveness of real-time SatRef corrections even on a smartphone-grade receiver.

4.5.2 Error Distribution and Scatter Plots

In addition to evaluating the accuracies of SPP and DGNSS, the SDAE was calculated to evaluate the precision. This can show the consistency of error magnitudes across the 22 survey stations. For the standard deviation of horizontal errors (σdPos), DGNSS is 31.7% less than SPP. Since DGNSS also has a lower MAE, this shows that DGNSS not only have lower error magnitudes, but the error distribution is also tighter. The standard deviation of vertical errors (σdH) for SPP (3.80m) was comparable to DGNSS (3.71m), a 2.3% decrease. To visualize both the accuracy and precision, scatter plots of positional errors were used, distributed around the center indicate higher accuracy, while clustered points indicate higher precision.

Figure 4E Scatter plot of horizontal position errors for SPP (red) versus DGNSS (blue) across the surveyed points.

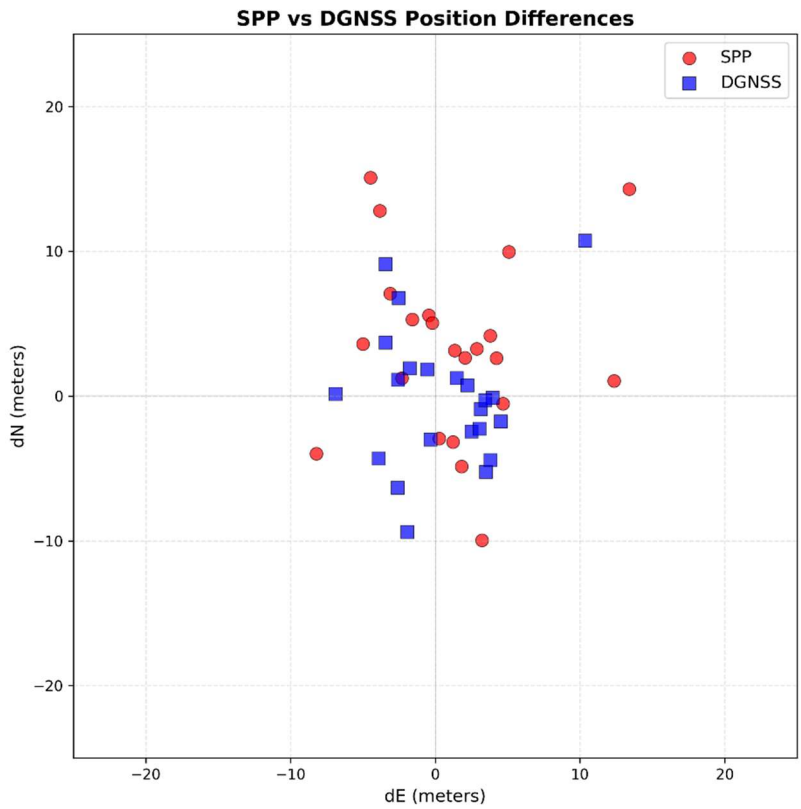


Figure 4E compares the horizontal position error between the final positions of SPP and DGNSS, plotting both the dE (x-axis) and dN (y-axis), visually analyzing the positional errors. SPP errors are represented by red dots, while DGNSS errors are represented by blue dots. They all cluster around the center (0, 0), against the coordinates of the corresponding control stations of each survey point, showing the surveyed points are generally accurate. Most points cluster are within 5m radius from the origin, which shows that SPP and DGNSS can usually achieve 5m accuracy. SPP dots appear to be more sparsely distributed, while DGNSS shows a tighter cluster and few outliers above 10m.

Figure 4F Scatter plot of vertical errors for SPP (red) versus DGNSS (blue) across the surveyed points.

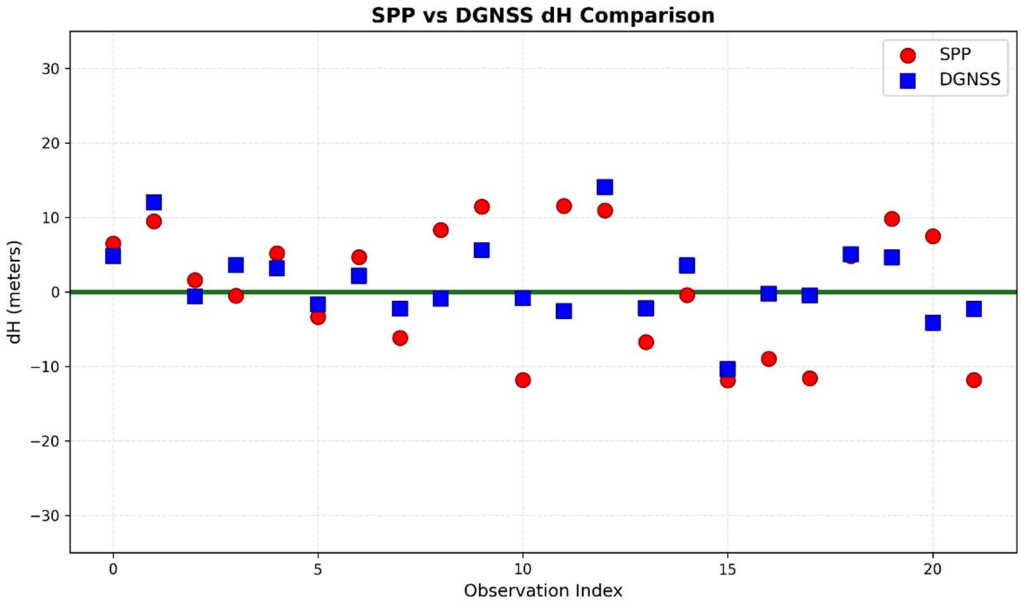


Figure 4F shows the vertical errors between the final positions of SPP and DGNSS, along the x-axis are the stations in arbitrary order, while along the y-axis are the vertical errors. In general, DGNSS clusters around the center line (0m error) more frequently, with some points having the relatively same error magnitude as SPP, therefore the MAE of

DGNSS is significantly lower, but the SDAE remains comparable to SPP, reflected by their similar variability.

4.6 Results Conclusion

The surveying results demonstrated the improvement of positional accuracy and horizontal precision of mobile DGNSS using SatRef service compared to standalone SPP. Among the 22 geodetic benchmarks, DGNSS showed improvement in 17, reduced horizontal MAE by 28.8% and vertical MAE by 36.3%, and improved the horizontal precision (lowered SDAE) by 31.7%. Both positioning methods can achieve meter-level accuracy, yet DGNSS outperformed SPP in 77% of the stations. These improvements prove the practicality of real-time corrections from SatRef services in Hong Kong, without the need to implement sophisticated atmospheric corrections on the receiver side, and still achieving a higher accuracy with simpler error modeling algorithm.

Chapter 5: Discussion

5.1 Effectiveness of DGNSS Correction Using SatRef

Firstly, the results in Chapter 4 has shown that DGNSS has generally outperformed standalone SPP. The horizontal MAE was reduced from 7.39m to 5.26m, improving 28.8%, and vertical MAE was reduced from 6.90m to 4.40m, improving 36.3%. These results match the expectation that DGNSS mitigates spatially correlated errors using calculated pseudorange correction is generally more reliable and accurate than SPP, which mitigates atmospheric delays using models.

Secondly, improvement is more notable in the vertical position (36.3% in MAE compared to 28.8% horizontally). This is likely due to the fact that atmospheric delays affect the vertical component more than horizontal, and the SatRef correction magnitude for vertical is likely larger. Since the SPP algorithm uses the Klobuchar ionospheric model and the Saastamoinen tropospheric model to estimate these delays, while DGNSS cancels them through differential correction, the correction magnitude and accuracy is likely larger, effectively reducing larger height error residuals compared to SPP.

Lastly, there is also notable improvement in the horizontal precision (SDAE reduction). DGNSS reduced the SDAE from 4.58m to 3.13m, a 31.7% reduction. As above mentioned, the MAE has also reduced. This shows that DGNSS is more accurate and consistent compared to SPP. This is also important for a positioning application, since consistent performance is often more practical than occasional high accuracy.

5.2 Stations where DGNSS did not improve

In 5 of the 22 surveyed stations (23%), DGNSS did not outperform SPP in either horizontal or vertical component. This is an expected outcome for several reasons.

Firstly, DGNSS cannot correct errors that are not spatially correlated between the base station and the rover. For instance, multipath effect is caused by signal reflections from buildings and structures near the smartphone. This is a local effect that varies across locations, and can be entirely different for the SatRef base station (clear-sky environment) and the rover (can encounter buildings, obstacles, humans). According to Medina et al. (2018), multipath can cause pseudorange biases of several meters in severe cases, and these biases are not cancelled by the DGNSS correction. In areas with dense buildings, such as Kowloon City district, is more susceptible to this effect.

Secondly, receiver noise is uncorrelated between the base station and the rover, furthermore, this project used Redmi 14C, which uses a low-cost smartphone antenna, the noise floor is often higher than a geodetic-grade receiver. The application used the Van Dierendonck (1996) code tracking jitter model as a weight function in the least squares adjustment to give less weight to noisier pseudoranges, but this cannot eliminate noise-induced errors.

Lastly, the measurement epochs might contain temporary systematic bias. Since Android devices have high clock drift, the clock bias should be solved epoch by epoch, and average the calculated positions. However, some stations are more susceptible to multipath or signal obstruction, for instance near a reflective glazed wall, the affected observations might pass the 2-sigma filter and skew the mean, which degrades the final position.

5.3 Impact of Smartphone Hardware on Absolute Accuracy

The absolute accuracy achieved was 5.26m horizontally and 4.40m vertically in MAE. However, according to the LandsD DGNSS Positioning Service (2026), the expected accuracy for DGNSS should be 1 to 2 meters, and the current implement is less accurate than LandsD claimed DGNSS is capable of. This is expected and is likely caused by the hardware limitations of Redmi 14C, rather than the DGNSS algorithm or the SatRef pseudorange corrections.

Firstly, Redmi 14C is a single-frequency receiver, tracking primarily the GPS L1 signal. According to Konno et al. (2006), dual frequency pseudoranges is the most effective method for eliminating first-order ionospheric delay. Therefore, single-frequency receivers cannot perform ionosphere-free linear combination. Although the Klobuchar model for SPP,

and differential correction for DGNSS both mitigate ionospheric error, residual ionospheric delay can still range up to several meters under high solar activity (Klobuchar, 1987).

Secondly, smartphone GNSS antennas are less sensitive than survey equipment grade antennas, the signal to noise ratio is usually lower, some measurements even had 0 dB (S/N = 1), indicating poor signal quality. From Figure 5A, around half of the measurements falls below 12 dB, and less than 1% exceed 25 dB. This shows that the noise is common in surveying, and adversely affects the measurement quality.

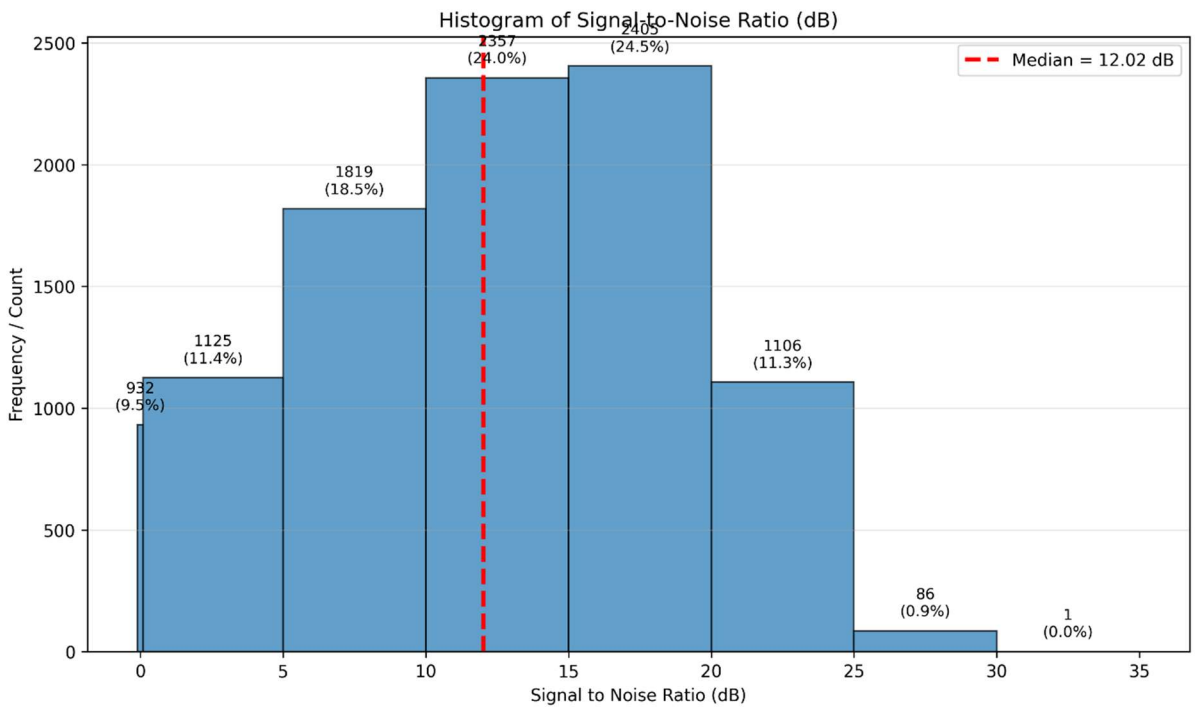


Figure 5A Histogram of signal to noise ratio dB of all observations

Even though this project uses a 5 dB threshold filter to mitigate noise, the measurements in general can still be noisy.

Thirdly, Android smartphones have significantly higher receiver clock drift compared to dedicated GNSS receivers. Instead of solving the same unknown receiver clock bias across close adjacent epochs, each epoch has to be solved independently, and take the average over

the epochs. However, the averaging process is sensitive to the quality and number of usable epochs, which varies with satellite visibility.

Despite the hardware constraints with a budget smartphone, the results still show that DGNSS with SatRef is effective and practical, since it is accurate to 5-meter level, it is practical for many location-based applications. Besides, the accuracy has improved significantly (MAE reduction), shows that there is meaningful improvement compared to SPP.

5.4 Adapted Source Code Components

This project handles GPS subframes and RTCM data, which involved adapting several existing open-source implementations.

5.4.1 GPS Subframe Decoding (adapted from Google GNSSLogger)

The Android API provides a class `GnssNavigationMessage`, the `data` field contains a subframe of a GPS satellite in byte array. The ephemeris parameters (e.g. semi-major axis, eccentricity) can be extracted from the byte array, according to the decoding bit positions and applying the scale factors specified in IS-GPS-200 (US DoD, 1995). This was adapted from Google's GNSSLogger project, which is written in Java, ported to Kotlin in accordance with the specifications.

5.4.2 RTCM v2 Word Decoding (adapted from RTKLIB rtcm.c)

This project decodes the RTCM in two stages, first is verifying the parity bits using the Hamming (32, 26) scheme and handling the bit-inversion rule, which is dependent on the D29 and D30 bits of the preceding word (RTCM, 1998), and extracts only the 24 bits of data, appends to the byte buffer. The second part is parsing the buffer as RTCM type 1 message. The first part was adapted from rtcm.c in RTKLIB, developed by Takasu (2013). The code has used many low-level bit manipulations.

5.4.3 RTCM v2 Type 1 Message Decoding (adapted from BKG RTCM2.cpp)

The second part of decoding the RTCM message involved extracting the PRC, RRC, UDRE, and satellite PRN from the continuous byte buffer. The parsing logic follows the bit layout specified by Betke (2001) and RTCM (1998), and was adapted from RTCM2.cpp, developed by the Federal Agency for Cartography and Geodesy (BKG, 2008). It has involved bit operations and extracting to signed and unsigned integers, and apply the correct scale factor to the PRC.

In all three sources, the adaptation process required a significant level of understanding of both the source implementation and the official specifications. Using established open-source implementation is standard practice in modern GNSS software development, avoids reinventing the wheel and can let the developers focus on validating the accuracy of DGNSS compared to SPP.

Chapter 6: Conclusion

This project has previously defined three objectives: to develop a mobile DGNSS positioning algorithm based on raw GNSS measurements and SatRef corrections, to assess the accuracy of DGNSS by comparing with geodetic ground truth and SPP, and to highlight possible limitations and suggest improvements. All three are accomplished within this thesis.

6.1 Development of Mobile DGNSS Algorithm

The application receives raw GPS navigation and observation messages, A-GNSS ephemeris through SUPL, and RTCM v2 DGNSS corrections from SatRef. It decodes binary files and implements pseudorange corrections and outputs to CSV/JSON files (so that any future modification can be done independently from resurveying) as well as does epoch-based least square solutions with weighted averaging. As for SPP, it uses Klobuchar and Saastamoinen models to correct the delay caused by atmosphere, whereas for DGNSS, pseudoranges are adjusted with PRC, and output to HK1980 Grid coordinate system to directly compare with the local geodetic control points. The code regarding parsing GPS and RTCM v2 messages were referenced and adapted from existing open-source codes. GPS subframe parsing from Google GNSSLogger (Google, 2023), RTCM v2 message parsing from RTKLIB (Takasu, 2013), and RTCM v2 Type 1 message parsing from BKG RTCM2.cpp. All were rewritten in Kotlin according to their corresponding specifications (US DoD, 1995; RTCM, 1998; Betke, 2001). Satellite position calculations and least squares adjustment are newly developed along with epoch averaging with 2-sigma rejection and coordinate transformation.

6.2 DGNSS Accuracy

DGNSS showed a considerable improvement in terms of accuracy and precision over standalone SPP for the geodetic benchmarks surveyed. In terms of accuracy, DGNSS has lower mean absolute error for both horizontal and vertical coordinates, and improves more vertically. This result is consistent with the differential correction removing the atmospheric delay bias affecting vertically more often. Horizontal position has also improved in consistency, showing a tighter distribution across the surveyed locations. Although both SPP and DGNSS achieved meter-level accuracy, with most errors below 10m, DGNSS showed results with higher accuracy and repeatability, which reflects SatRef corrections for smartphone positioning in Hong Kong is effective and practical.

6.3 Limitations

In this project, there were several factors that limited the accuracy, most of which were caused by the smartphone hardware, and positioning algorithms to a lesser extent.

Firstly, Redmi 14C is a single-frequency receiver, thus it is unable to take ionosphere-free pseudorange measurements for SPP. In the algorithm, the Klobuchar ionospheric model was used for mitigation, but it only partially removed the ionospheric delay. This might leave some ionospheric delay residuals when comparing DGNSS against SPP, especially when there is high solar activity.

Secondly, only GPS navigation messages were used, and this could reduce the number of possible satellites used. Since each satellite gives one equation to the least squares adjustment, having fewer satellites reduces the redundancy and the final position accuracy.

Thirdly, smartphone grade GNSS antennas experience higher noise compared to dedicated GNSS survey instrument. Higher noise results in poorer measurements, and although the algorithm filters out the measurements with low signal to noise ratio, and uses

Van Dierendonck jitter model for the weight matrix, the measurements in general are noisier, and thus less accurate.

Fourthly, Android smartphones have high receiver clock drifts, thus it is infeasible to use multiple observation epochs in the same least squares. The current implementation solves the receiver clock bias for each epoch, averaging all the resulting positions. The trade-off is that individual epochs have less redundancy. Even with 2-sigma rejection, averaging the positions cannot remove long-term systematic bias, and still keeps the suboptimal measurements, contributing to the average position bias.

Lastly, SPP used fixed values for meteorological parameters, while DGNSS effectively "cancels" the tropospheric delays. Without real-time meteorological data, the actual weather condition might deviate from the assumed constants, increasing the residual tropospheric delay when comparing DGNSS with SPP.

6.4 Recommendations for Future Work

Taking into account the limitations discussed above, the following improvements are suggested for future development.

Firstly, real-time meteorology data from the Hong Kong Observatory API should be used. The current implementation uses constants based on typical weather conditions in Hong Kong, while in reality, weather changes from day to day and increases the tropospheric delay residuals. With the Saastamoinen model using real world data, including temperature, humidity, and air pressure, the accuracy of SPP would likely increase.

Secondly, support for other satellite constellations such as GLONASS and BeiDou should be used. This would help increase the number of satellites available for observation,

higher redundancy for least squares and higher positional accuracy. Therefore, it is believed that this change is the most effective improving the positional accuracy and applies to both SPP and DGNSS.

Lastly, a smartphone capable of receiving dual-frequency GNSS measurements should be used. This would enable ionosphere-free SPP and increase the accuracy compared to using Redmi 14C. Alternatively, using a smartphone with higher quality GNSS antenna could also improve the accuracy for single-frequency SPP.

6.5 Closing Statement

This project shows that accurate meter-level positioning is achievable on a consumer Android smartphone using real-time SatRef DGNSS corrections, and DGNSS provide better accuracy over SPP for most locations in Hong Kong. While the positioning accuracy still falls short compared to dedicated GNSS surveying equipment, this result still shows that SatRef provides effective assistance obtaining accurate positions for mobile phone users. The navigation messages, observations, and DGNSS corrections are stored in common file formats can be easily reused to verify the algorithm and accuracy.

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Appendix

1. Application source code (GitHub repository link)

https://github.com/Waterdragen/FinalYear

2. Raw Surveyed Files (OneDrive link to CSV/JSON files)

https://connectpolyu-my.sharepoint.com/:f:/g/personal/24001355d_connect_polyu_hk/IgDtVKU-rzpRRoClJbwgUcyYAYCbZP4jB-FOBG3gLwCJuAU?e=gawCff